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Hyemi Kim (she/her)

Research Interests

Socially Responsible Operations
Pricing and data-driven optimization, with applications in urban mobility and energy systems

Education

Aug. 2021 – present **Ph.D. in Operations Research (expected May 2027)**, *Columbia University*, New York, USA
Advisor: Prof. Adam N. Elmachtoub

Sept. 2018 – Aug. 2020 **M.S. in Industrial & Systems Engineering**, *KAIST*, Daejeon, Korea
Advisor: Prof. Il-Chul Moon

Feb. 2012 – Aug. 2018 **B.S. in Industrial & Systems Engineering and Electrical Engineering (double major)**, *KAIST*, Daejeon, Korea
Summa Cum Laude, Graduated first in the Department of Industrial & Systems Engineering

Research

(α-β) alphabetical author order * equal contribution

[1] **Fair Aggregation in Virtual Power Plants**
Liudong Chen*, Hyemi Kim*, Adam N. Elmachtoub, Bolun Xu
(Non-Archival) Conference on Fairness, Accountability, and Transparency (FAccT), 2026.

[2] **Learning Fair Demand Models**
Adam N. Elmachtoub, Hyemi Kim, Jonathan Y. Tan (α-β)
Major Revision at Operations Research
(Non-Archival) Conference on Fairness, Accountability, and Transparency (FAccT), 2026.

[3] **Fair Fares for Vehicle Sharing Systems**
Adam N. Elmachtoub and Hyemi Kim (α-β)
Operations Research, 2026.
(Non-Archival) Conference on Fairness, Accountability, and Transparency (FAccT), 2025.
- *Finalist, INFORMS DEI Best Student Paper Award, 2024*
- *Finalist, INFORMS Transportation Science and Logistics Society Best Student Paper Award, 2024*

[4] **Black-Box EM Algorithm for Estimating Latent States of High-Speed Vehicles**
Yoon-Yeong Kim, Hyemi Kim, WonSung Lee, Han-Lim Choi, and Il-Chul Moon
AIAA SciTech Forum, 2021.

[5] **Counterfactual Fairness with Disentangled Causal Effect Variational Autoencoder**
Hyemi Kim, Seungjae Shin, JoonHo Jang, Kyungwoo Song, Weonyoung Joo, Wanmo Kang, and Il-Chul Moon
Association for the Advancement of Artificial Intelligence (AAAI), 2021.

[6] **Neutralizing Gender Bias in Word Embedding with Latent Disentanglement and Counterfactual Generation**
Seungjae Shin, Kyungwoo Song, Joonho Jang, Hyemi Kim, Weonyoung Joo, and Il-Chul Moon
Findings of Empirical Methods in Natural Language Processing (EMNLP), 2020.

[7] **Deep Generative Positive-Unlabeled Learning under Selection Bias**
ByeongHu Na, Hyemi Kim, Kyungwoo Song, Weonyoung Joo, Yoonyeong Kim, and Il-Chul Moon
Conference on Information and Knowledge Management (CIKM), 2020.

Teaching

Teaching Assistant

Mean overall quality rating out of 5.0 in parentheses, where available

Fall 2024 Introduction to Probability and Statistics, Columbia IEOR (4.67)
Spring 2024 Stochastic Models, Columbia IEOR (4.64)
Fall 2023 Analytics in Action, Columbia Business School
Spring 2023 Transportation Analytics & Logistics, Columbia IEOR (4.60)
Spring 2022 Causal Inference, Columbia IEOR (4.40)
Fall 2019 Data Structures, KAIST ISE
Fall 2018 Data Structures, KAIST ISE

Pedagogical Training

2026 The MBA Transformed? AI and Beyond, Columbia Business School
2024 Professional Presentation, Columbia University

Honors and Awards

2025 Columbia-Dream Sports AI Innovation Fellowship, Columbia University
2024 Finalist, INFORMS DEI Best Student Paper Award
2024 Finalist, INFORMS Transportation Science and Logistics Society Best Student Paper Award
2023 Finalist, INFORMS BSS Data Challenge Competition
Report: Decoupling Price and Demand Estimation for Profit Optimization
2023 Deming Doctoral Fellowship, Columbia University
2019 Songhyun Award, KAIST
2012 Presidential Science Scholarship, Korea Student Aid Foundation (KOSAF)

Academic Service

2024 – 2026 Program Committee, Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)
2025 – 2026 Program Committee, IJCAI Special Track on AI and Social Good
2026 Session Chair, INFORMS Annual Meeting
2026 Session Chair, ACM Conference on Fairness, Accountability, and Transparency (FAccT)
2026 Reviewer, ACM Conference on Fairness, Accountability, and Transparency (FAccT)
Reviewer, Socio-Economic Planning Sciences
Reviewer, Journal of Risk and Insurance

Industry Experience

Jun. 2024 – Aug. 2024 **Data Scientist Intern**, *MOLOCO, Inc.*, Redwood City, USA
Designed and executed budget pacing simulations for online advertisements.

Jan. 2021 – Jul. 2021 **AI Research Intern**, *NAVER Corporation*, Seongnam, Korea
Studied test-time training methods for robust machine learning.

Jun. 2015 – Aug. 2015 **Intern**, *Samsung Electronics*, Suwon, Korea
Analyzed smart car development trends and improved patient analysis algorithms.

Talks

Fair Fares for Vehicle Sharing Systems

2026 INFORMS Annual Meeting, San Francisco, CA
2025 ACM FAccT Conference, Athens, Greece
2025 Invited Seminar, KAIST, Daejeon, Korea
2025 INFORMS TSL Workshop, Seoul, Korea
2024 INFORMS Annual Meeting, DEI Best Student Paper Award Session, Seattle, WA
2024 INFORMS Annual Meeting, TSL Best Student Paper Award Session, Seattle, WA
2024 INFORMS RMP Conference, Los Angeles, CA
2024 Market Innovation Workshop, Virtual
2023 INFORMS Annual Meeting, Phoenix, AZ
2023 Workshop on AI & Analytics for Social Good, College Park, MD

Learning Fair Demand Models

2025 INFORMS Annual Meeting, Atlanta, GA
2025 INFORMS RMP Conference, New York, NY

Fair Aggregation in Virtual Power Plants

2026 INFORMS RMP Conference, Ann Arbor, MI
2026 ACM FAccT Conference, Montreal, Canada
2026 Market Innovation Workshop, Virtual

Skills & Personal

Programming Python, MATLAB, and C
Tools PyTorch, TensorFlow, Gurobi, and Pyomo
Languages Korean (native), English (professional)
Hobby Brazilian jiu-jitsu (purple belt)

References

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